Which workflows are red, since when, and why?

EVERRED · GREEN→RED · RED→GREEN across the five federation core repositories, plus a gap analysis of what our own tooling can and cannot answer.

5 core repositories traced	**340** workflows with history	**16,008** runs read (GET only)	**43** red now · fresh
26 red · stale (replay first)	**2** red on pinned evidence	**10** shared break roots	**6/13** gaps filled / exceeded

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What are the core repos? Pipeline vs federation

Pipeline

The 'pipeline' is pipeline-automation-hub. It began as the Master_Input document pipeline (PPTX/DOCX → digital twins, PROCESSING_MANIFEST.json, last run 2025-09-10) and grew into Mission Control, GOGREEN, CODE PURPLE and JADE: the federation's read-only control plane.

Federation core (5)

- **pipeline-automation-hub**: Mission Control hub / control plane / Pages publisher
- **ABACUS**: System Under Test; EVERRED ledger owner; local tree named Master_Input
- **CODEX**: Federation governance & orchestration (Mycelium, receipts)
- **cryoplant-project**: QPS engineering SSOT and TRIAGE owner (private)
- **gg_MATH**: Mathematics provider (PCA/BT/MC kernels)

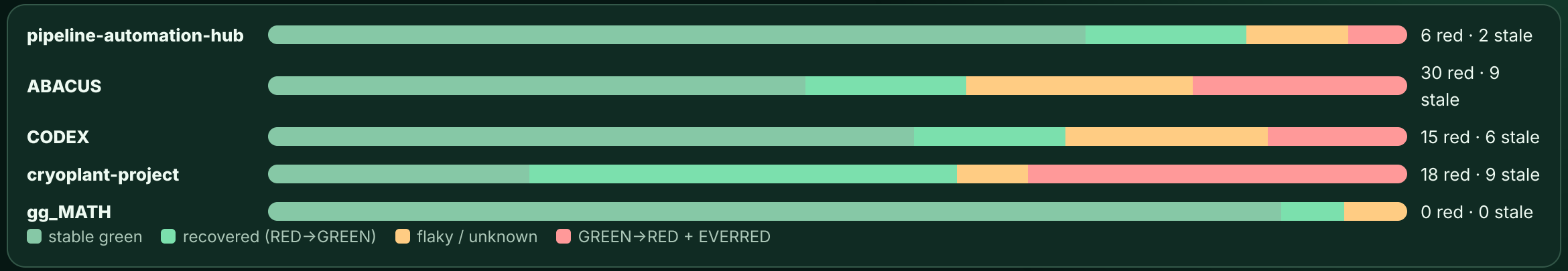
Satellites & cohort

DOCX_RTM_Automation, Q_engineering_tools, document-organization-system

TOP21 roster = 5 core + 3 satellites + 13 others, of which 9 are upstream forks (ariana, cli, docker-stacks, github-mcp-server, prettier, python-sdk, skills, vscode-powershell, yaml-grammar).

JADE N+8 · RED-TRACE

Federation scoreboard



Red now: **43** fresh, **26** stale (replay before trusting), **2** on pinned historical evidence.

JADE N+8 · PIPELINE-AUTOMATION-HUB

pipeline-automation-hub: 6 red · 2 stale

Mission Control hub · pipeline origin (Master_Input document pipeline → MC/JADE/GOGREEN)

WORKFLOW	CLASS	FLAGS	WINDOW	FIRST FAILING STEP
GM-IV RECON_4 Governor Gate	GREEN→RED	RED NOW	3a35154a → 612dd404 (4 red)	recon4-governor: Execute RECON_4 Governor gate
GM-V Governor Readiness	GREEN→RED	RED NOW · 📌 PINNED EVIDENCE	445bced6 → 127f84b7 (2 red)	gm-v-governor-readiness: Verify exact GM-IV runtime capacity run and artifact identity
pages build and deployment	GREEN→RED	RED NOW · STALE 4.5 d · * history truncated	a3fb9a5a → 3b468f1e (1 red)	deploy: Deploy to GitHub Pages
Mission Control N3 Evidence Metrics	EVERRED	RED NOW	d1012487	N3 evidence Python 3.12: Run bounded N3 unit, adversarial and CLI tests; N3 evidence Pytho
GM-V 3PSTAR MIP Control	FLAKY	RED NOW · STALE 31.0 d · 📌 PINNED EVIDENCE	75986e49 → 6980c60f (2 red)	gm-v-3pstar-mip-control: Compile and run deterministic 3PSTAR MIP validator
Zero-Test Gate Guard	FLAKY	RED NOW · * history truncated	76bafe29 → c3b874fe (1 red)	audit: Every unittest gate collects at least one test

JADE N+8 · ABACUS

ABACUS: 30 red · 9 stale

System Under Test · 12-cluster DMAIC engine · EVERRED ledger owner (local working tree: Master_Input)

WORKFLOW	CLASS	FLAGS	WINDOW	FIRST FAILING STEP
OSV-Scanner Scheduled	GREEN→RED	RED NOW	599b6055 → d3db3141 (4 red)	OSV Scan (schedule / manual) / osv-scan: Check that the scan completed, Upload to code-sca
TestPilot Pre-GitHub Parity	GREEN→RED	RED NOW	5f0dfebb → 2e8e8b95 (11 red)	parity: Run bounded TestPilot fast preflight
.github/workflows/qps-debug-dow-consumer.yml	EVERRED	RED NOW STALE 37.6 d	1b1366e6	
ABACUS DOW Integration CI/CD	EVERRED	RED NOW STALE 23.9 d ⁺ history truncated	a0de73a9	Test DOW Phases (DMAIC): Run DOW Phase Tests
Auto-Merge PRs	EVERRED	RED NOW STALE 51.4 d ⁺ history truncated	505e0fb4	
Automatic Dependency Submission (Python)	EVERRED	RED NOW STALE 110.6 d ⁺ history truncated	799ec6a3	submit-pypi: validate-project
Bandit Disjoint Census	EVERRED	RED NOW STALE 5.9 d	02cca502	census: Check committed governance inventory, Validate coverage and JUnit receipts

+23 more on the HTML page.

JADE N+8 · CODEX

CODEX: 15 red · 6 stale

Federation governance & orchestration hub (Mycelium, MissionControl receipts)

WORKFLOW	CLASS	FLAGS	WINDOW	FIRST FAILING STEP
MASTER Contract SSOT Validation	GREEN→RED	RED NOW	c552769a → 855acb78 (5 red)	validate-master-contract-ssot: Validate MASTER contract SSOT
SPINE Gate	GREEN→RED	RED NOW	b75fc4e7 → 18b5d47b (1 red)	spine-gate: Receipt must be regenerated in the same PR
.github/workflows/qps-debug-federation-consumer.yml	EVERRED	RED NOW, STALE 37.5 d	b7bc79ea	
.github/workflows/release.yml	EVERRED	RED NOW, STALE 142.5 d	025ab083	
.github/workflows/w70-qps-zero-delta-diagnostic.yml	EVERRED	RED NOW, STALE 27.7 d, ⁺ history truncated	1bc865a0	
G5 Unified Lifecycle Execution & Deployment Gateway	EVERRED	RED NOW, STALE 141.0 d	3dbd0662	g5-build-and-deploy: Execute G5 Master Orchestrator and Bake Artifacts
OSV-Scanner	EVERRED	RED NOW, ⁺ history truncated	b6421d7f	scan-scheduled / osv-scan: Run osv-scanner-reporter

+8 more on the HTML page.

JADE N+8 · CRYOPLANT-PROJECT

cryoplant-project: 18 red · 9 stale

QPS engineering SSOT · QPS TRIAGE owner (private)

WORKFLOW	CLASS	FLAGS	WINDOW	FIRST FAILING STEP
QPS BD Queue Integrity	GREEN→RED	RED NOW	666d7766 → 4782b8ab (1 red)	parity: Validate live issue parity and partial-return no-close guards
QPS Dependency SPINE Read-only Census	GREEN→RED	RED NOW	e72e4454 → 152d7b4b (12 red)	Five-repository contract, QPS exact-head native observer:
QPS PR binary refresh	GREEN→RED	RED NOW, * history truncated	b3ac5041 → 8102bce3 (14 red)	refresh-open-prs-after-main:
QPS TRIAGE Debug DAG Pilot	GREEN→RED	RED NOW, * history truncated	0c77a6bc → 9caccb14 (3 red)	plan DAG and prove session/port contract: ; roll up debug DAG metrics and next pulse:
TODO-011 release pack smoke	GREEN→RED	RED NOW, STALE 33.1 d, * history truncated	cf0f7137 → aeb16b22 (33 red)	release-smoke:
TRIAGE Repository Census	GREEN→RED	RED NOW, * history truncated	b3ac5041 → 8102bce3 (6 red)	runner-preflight:
github_actions in /. - Update #1630350736	GREEN→RED	RED NOW, STALE 4.3 d	a5fc3eaf → 59315bbb (1 red)	Dependabot: Run Dependabot

+11 more on the HTML page.

gg_MATH: 0 red · 0 stale

Mathematics provider (PCA/BT/Monte Carlo kernels, Plotly artifacts)

No workflow red at last observation. Recovered episodes are listed on the HTML page.

Gap analysis (1/3): what our own code could not answer

AGENT PARTIAL **G01 · One question, four answers**

Asking 'which repos are core' gives 4 different lists (TOP21 roster 21, TOP7 census 7, live_census 5, W1 rebuild 5). TOP21 includes 9 forks.

Fill: Crosswalk computed from all 4 SSOTs + live org (this iteration).

Next: Owner: name ONE canonical registry; others derive.

AGENT FILLED **G02 · Red-trace saw half the fleet**

Tracer listed only the first 100 workflows; ABACUS has 244, CODEX 174, QPS 149, hub 162.

Fill: Paginated listing + fail-closed if census incomplete; 2 regression tests.

Next: —

AGENT EXCEEDED **G03 · Hub-only measurement**

Ledger covered 1 of 5 core repos.

Fill: Same tracer run over all 5 core repos; federation snapshot drives HTML/slides/PDF.

Next: Schedule federation trace (needs cross-repo read token for private QPS in CI).

AGENT PARTIAL **G04 · Stale reds never re-evaluated**

Path-filtered workflows keep a red verdict for weeks.

Fill: 3 replays dispatched: m3-lossless and GM-IV capacity proof v2 flipped to GREEN; RECON_4 stays red (superseded gate).

Next: Owner: approve bounded auto-replay lane (actions:write).

OWNER DETECTED **G05 · Retention time-bombs**

2 GM-V gates assert on purged artifacts (30-day retention); red with zero code change.

Fill: 📌 marker + PINNED_HISTORICAL_EVIDENCE cause in ledger; fresh native evidence now exists (run 39293734282, expires 2026-11-18).

Next: Owner: bind to latest valid run, not a pinned id.

OWNER DETECTED **G06 · Superseded stage gates**

GM-IV RECON_4 gate rejects the later ACTIVE_8 state → permanent red.

Fill: Root traced (mode=INVALID).

Next: Owner: retire, or classify later stage as SUPERSEDED.

Gap analysis (2/3): what our own code could not answer

AGENT FILLED **G07 · Required checks would deadlock**

Method Atlas & Live Lossless are path-filtered; requiring them blocks unrelated PRs forever.

Fill: Always-reporting GOGREEN Required Gate aggregates them on exact head (13 tests).

Next: Owner: require only 'GOGREEN required gate' on master.

AGENT FILLED **G08 · Two ledgers, no join**

Hub red-trace and ABACUS EVERRED ledger did not share events.

Fill: EVERRED ingest_red_trace adapter (5 tests, EVERRED 376/376) + events appended to Master_Input/ABACUS ledger.

Next: —

OWNER OPEN **G09 · Master_Input pipeline frozen**

Document pipeline manifest last generated 2025-09-10; scan route hard-codes file names.

Fill: Recorded; not changed (out of scope, live app).

Next: Owner: re-run or retire the document pipeline.

AGENT PARTIAL **G10 · Opaque failures**

QA steps reported only 'exit code 1'.

Fill: Annotation wrapper on W1/W2/Live Lossless (N+0..N+7).

Next: Extend wrapper to remaining QA steps.

AGENT FILLED **G11 · Implicit agent involvement**

Pages did not show which actions an agent took vs owner/CI.

Fill: Explicit actor badges (AGENT · OWNER · CI) on every action/gap in HTML, slides and PDF.

Next: —

AGENT FILLED **G12 · No shareable evidence pack**

Ledger only in JSON/markdown.

Fill: JADE HTML page + 16:9 HTML slides + interactive PDF (bookmarks, internal & source links).

Next: —

Gap analysis (3/3): what our own code could not answer

OWNER OPEN **G13 · Pages publication contract**

The hosted-parity contract pins JADE's exact public asset set; adding the red-trace child changes the published Pages contract.

Fill: Red-trace HTML/slides/PDF built and validated as a JADE child; staging deliberately not wired into portal_bridge.

Next: Owner: approve extending portal_bridge + hosted_parity asset set (one follow-up PR).

Explicit agentic involvement

AGENT executed · OWNER decided · CI evidence. Agents never merge, waive or confirm roots.

DATE	ACTOR	ACTION	REFERENCE
2026-10-18	AGENT	Repaired 8 master first-red roots	PRs #698 #699 #702 #703 #705 #707 #720
2026-10-18	OWNER	Merged all repairs	#698–#720
2026-10-19	AGENT	Built red-trace ledger	#752
2026-10-19	OWNER	Merged red-trace ledger	#752
2026-10-19	AGENT	Dispatched 3 stale-red replays (owner-authorised 'do all unblocked')	m3-lossless ✓, capacity proof v2 ✓, RECON_4 ×
2026-10-19	AGENT	Root-traced RECON_4 (superseded stage) and GM-V (purged artifacts)	#529
2026-10-20	AGENT	Fixed tracer workflow pagination; federation trace ×5 repos	this PR
2026-10-20	AGENT	Built GOGREEN required-gate aggregator	this PR
2026-10-20	AGENT	EVERRED adapter + ledger append	ABACUS PR
2026-10-20	CI	Native verification of every PR head	Actions
—	OWNER	Branch protection, evidence rebind, gate retirement, replay lane	pending decisions

Owner decisions that unblock the rest

1. **G01** One question, four answers: Owner: name ONE canonical registry; others derive.
2. **G04** Stale reds never re-evaluated: Owner: approve bounded auto-replay lane (actions:write).
3. **G05** Retention time-bombs: Owner: bind to latest valid run, not a pinned id.
4. **G06** Superseded stage gates: Owner: retire, or classify later stage as SUPERSEDED.
5. **G09** Master_Input pipeline frozen: Owner: re-run or retire the document pipeline.
6. **G10** Opaque failures: Extend wrapper to remaining QA steps.
7. **G13** Pages publication contract: Owner: approve extending portal_bridge + hosted_parity asset set (one follow-up PR).

Sources & replay

- GBOGEB/pipeline-automation-hub: 78 workflows, 2,974 runs, digest `121fae09ff31`
- GBOGEB/ABACUS: 106 workflows, 6,808 runs, digest `db6452f1dcc2`
- GBOGEB/CODEX: 90 workflows, 4,655 runs, digest `90b7fb65ceca`
- GBOGEB/cryoplant-project: 48 workflows, 1,307 runs, digest `0f6ba1c16af2`
- GBOGEB/gg_MATH: 18 workflows, 264 runs, digest `ab32cfda5a93`
- Replay: `python mission-control/gogreen/red_trace/red_trace.py --repo … --root … --out …` then `tools/build_redtrace_data.py` and `site/build_redtrace.py`